

Statistical Physics of Particles – Mehran Kardar

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1 Thermodynamics

1.1 Laws of Thermodynamics

0. If systems A, B are each in equilibrium with system C , then A and B are also in equilibrium with each other.
1. Let E be the **internal energy** of the system. In adiabatic changes, $dE = \delta W$. If the change is not adiabatic,

$$dE = \delta W + \delta Q$$

δQ is the **heat intake** of the system. δQ and δW are inexact differentials (depend on path) while dE is an exact differential (depend only on endpoints). Note that $\delta Q, \delta W$ are positive if energy is going into the system.

2. (Kelvin) A process whose sole result is the complete conversion of heat to work is not possible. (Clausius) A process whose sole result is the transfer of heat from a colder to hotter body is not possible.
3. The entropy of all systems at 0 absolute temperature is a universal constant that can be taken to 0.

$$\lim_{T \rightarrow 0} S = 0$$

*The 0th law makes it possible to have a well-defined equilibrium state.

*The 1st law is just conservation of energy.

*The 2nd essentially says: heat flows naturally from hot to cold.

- Heat engines take Q_H from hot reservoir, extract work W , and dump remainder Q_C into cold reservoir. Efficiency is

$$\eta = \frac{W}{Q_H} = \frac{Q_H - Q_C}{Q_H} \leq 1$$

- Refrigerators take in W , pull Q_C from cold reservoir, and dump resulting Q_H into hot reservoir.

$$\omega = \frac{Q_C}{W} = \frac{Q_C}{Q_H - Q_C}$$

- Kelvin's statement rules out perfect heat engines. Clausius's statement rules out perfect refrigerators.
- By juxtaposing an ideal heat engine + fridge or ideal fridge + heat engine, we can show that Kelvin and Clausius's statements are equivalent.

1.2 Quasi-Static Transformations

Quasi-static transformation: By changing the system slowly enough, we can ensure the system is at equilibrium at every stage.

Adiabatic transformation: $\delta Q = 0$

Let the state function be $\{X_1, X_2, \dots, X_n\}$ and the generalized displacements be $\{x_1, x_2, \dots, x_n\}$. Define the conjugate generalized forces $\{F_1, F_2, \dots, F_n\}$. In an infinitesimal quasi-static transformation,

$$\delta W = \sum_i J_i dx_i$$

E.g. The force can be $-P$ and the displacement is V . Negative force, because by increasing force, V decreases. Cf. Increasing tension T on wire increases displacement L .

Heat capacities are changes in temperature upon addition of heat to the system. They depend on the path:

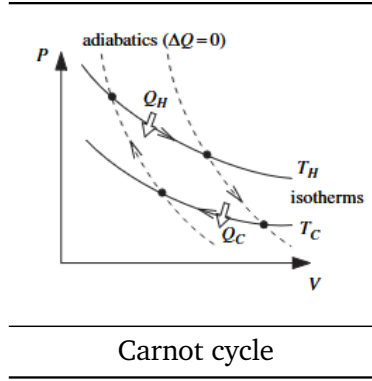
$$C_V = \left. \frac{\delta Q}{dT} \right|_V = \left. \frac{dE - \delta W}{dT} \right|_V = \left. \frac{dE + PdV}{dT} \right|_V = \left. \frac{\partial E}{\partial T} \right|_V$$

$$C_P = \left. \frac{\delta Q}{dT} \right|_P = \left. \frac{dE - \delta W}{dT} \right|_P = \left. \frac{dE + PdV}{dT} \right|_P = \left. \frac{\partial E}{\partial T} \right|_P + P \left. \frac{\partial V}{\partial T} \right|_P$$

1.3 Carnot Engines

Carnot engine: A reversible engine that runs in a cycle. All heat exchanges happen at source temperature T_H and cold sink T_C .

The Carnot engine is adiabatic ($\delta Q = 0$) except at reservoirs, where it's an isotherm. Assume a monoatomic gas with ideal energy $E = \frac{3}{2}NkT = \frac{3}{2}PV$. The $\delta Q = 0$ condition implies $PV^\gamma = \text{const.}$ in adiabatic paths, with $\gamma = 5/3$ for ideal gas.



Carnot's Theorem: The Carnot engine is the most efficient engine running between two reservoirs.

- Can be proven by running arbitrary heat engine in tandem with reversed Carnot fridge.
- The efficiency of a Carnot engine is

$$\eta = \frac{T_H - T_C}{T_H}, \quad \frac{Q_H}{Q_C} \equiv \frac{T_H}{T_C}$$

1.4 Functions of State

1.4.1 Entropy

Clausius's Theorem: For any cyclic transformation,

$$\oint \frac{\delta Q}{T} \leq 0$$

Where δQ is the heat supplied to the system at temperature T .

For a reversible process, $\oint \delta Q/T = 0$ by splitting the cycle into two paths of δQ_{rev} and $-\delta Q_{rev}$. We construct the function of state **entropy** for reversible processes:

$$\Delta S_{A \rightarrow B} = \int_A^B \frac{\delta Q_{rev}}{T}$$

- Thus for any reversible cycle, the heat is computed as $\delta Q_{rev} = TdS$. The 1st law becomes

$$dE = \delta Q + \delta W = TdS + \mathbf{F} \cdot d\mathbf{x}$$

- For irreversible processes from A to B , construct a reversible path from B to A to get

$$\int_A^B \frac{\delta Q}{T} + \int_B^A \frac{\delta Q_{rev}}{T} \leq 0$$

$$\int_A^B \frac{\delta Q}{T} \leq S(B) - S(A) = \Delta S_{A \rightarrow B}$$

Thus $dS \geq \delta Q/T$.

- If the system is **adiabatic**, $\delta Q = 0$ and $dS \geq 0$. Internal changes can only increase S , so S is maximum at equilibrium.
- The direction of increasing entropy provides an arrow of time.

1.4.2 Enthalpy

Previously, **entropy** was maximized for an adiabatic system in equilibrium. **Enthalpy** is a state function that's maximized in a system subject to a constant external force.

For work input δW , the displacement satisfies $\delta W \leq \mathbf{F} \cdot d\mathbf{x}$. In an adiabatic system, $\delta Q = 0$ and $dE \leq \mathbf{F} \cdot d\mathbf{x}$. Define the enthalpy H so that

$$\delta H \leq 0, \quad H = E - \mathbf{F} \cdot d\mathbf{x}$$

1.4.3 Helmholtz Free Energy

Applies to isotherms with $\delta W = 0$. From Clausius's theorem, $\delta Q \leq TdS$. If $\delta W = 0$, then $dE \leq TdS$. Define the **Helmholtz free energy** F as

$$\delta F \leq 0, \quad F = E - TS$$

1.4.4 Gibbs Free Energy

Applies to isotherms with const. mechanical force. From $\delta W \leq \mathbf{F} \cdot d\mathbf{x}$ and $\delta Q \leq TdS$, we have $dE \leq TdS + \mathbf{F} \cdot d\mathbf{x}$. Define **Gibbs free energy** G such that

$$\delta G \leq 0, \quad G = E - TS - \mathbf{F} \cdot \mathbf{x}$$

Note that $\Delta G = \Delta H - T\Delta S$, and indeed this is used in chemistry to determine whether a reaction is spontaneous.

1.4.5 Grand Potential

Chemical work can be involved if the particle number isn't conserved.

$$\begin{aligned} \delta W &= \vec{\mu} \cdot d\vec{N} \\ \vec{\mu} &= (\mu_1, \mu_2, \dots, \mu_n) \\ \vec{N} &= (N_1, N_2, \dots, N_n) \end{aligned}$$

μ_i are the chemical potentials necessary to add particle of species i to the system. N_i are the number of particles of species i .

This is treated separately from mechanical work and doesn't appear in Gibbs free energy. For chemical equilibrium + no mechanical work, the **grand potential** is

$$\begin{aligned} \mathcal{G} &= E - TS - \vec{\mu} \cdot \vec{N} \\ d\mathcal{G} &= -SdT + \mathbf{F} \cdot d\mathbf{x} - \vec{N} \cdot d\vec{\mu} \end{aligned}$$

In summary:

$\delta Q = 0$ $T = \text{const.}$		
$\delta W = 0$	$\delta S \geq 0$	$\delta F \leq 0$
$\mathbf{F} = \text{const.}$	$\delta H \leq 0$	$\delta G \leq 0$